

Monologue on Operating Systems

Step 1. Introduction

Linus Torvalds, the creator of Linux, once said: “Talk is cheap. Show me the code.” I like this quote because it reminds us that behind every computer there is an **operating system** that really does the work. Today, every computer, phone, or smart watch needs an OS to run properly. *Firstly*, I will explain what an operating system is and what types of OS exist. *Moreover*, I will talk about what operating systems actually do. *Finally*, I will discuss the history of OS and compare two modern ones.

Step 2. What is an OS?

2.1. *To begin with*, an **operating system** is a **software on the hard drive** that **enables the computer hardware to communicate and operate with the computer software**. Without it, the computer would simply be a useless box of metal and plastic.

2.2. There are several types of OS, and one system often belongs to more than one type. *Firstly*, a **GUI operating system** contains **graphics and icons** and is navigated with a mouse — Windows 11 and macOS **use a graphical user interface**. *Furthermore*, a **multi-user operating system allows for multiple users to use the same computer at the same time and at different times**, like Linux or Unix servers. *In addition*, a **multiprocessing operating system can support and utilize more than one computer processor**. *On the one hand*, a **multitasking operating system can allow multiple software processes to run at the same time**. *On the other hand*, a **multithreading operating system can allow different parts of a software program to run concurrently**.

Step 3. What do OS do?

3.1. The operating system is one of the most important parts of a computer because it has to **control the computer's/system resources** — the processor, memory, and disk space. When choosing an OS for business, people look at the hardware platform, security, the ease of administration, and **the adaptability towards different uses**. *Moreover*, an OS has to **provide a consistent application program interface (API)**, so software written for one machine can run on others.

3.2. Operating systems **accomplish** several key tasks. *For instance*, the OS has to **allocate resources and allocate enough of the processor's time to each process and application**, so that everything can **run efficiently and properly** when users **have multiple applications and processes running**. *Furthermore*, the OS must **ensure that each process has enough memory to execute the process**, and **balance the needs of each process with the different types of memory available**. *Additionally*, drivers **translate the electrical signals sent from the operating system or application**

program to the hardware device. *As a result,* the OS also provides the user interface for people to work with their computer.

Step 4. History of OS

4.1. Operating systems have evolved through several generations. At first, the earliest computers had no OS at all — programmers had to **enter a program one bit at a time on rows of mechanical switches.** *However,* in the early 1950s, the first **single-stream batch processing systems** appeared; they could only **run one job at a time**, and people had to **submit programs and data in groups/batches.** *Later,* in the 1960s, engineers **developed the concept of multiprogramming** to **take better advantage of the computer's resources** and to **run several jobs at once** by **switching from job to job as needed, keeping several jobs advancing while keeping the peripheral devices in use.** *Furthermore,* they **developed a time-sharing technique** and a **spooling technique** to **multiprogram a large number of simultaneous interactive users.**

4.2. As for modern OS, I use Windows 11 with WSL Ubuntu, because I work in QA automation and need both worlds. *On the one hand,* Windows is user-friendly and is **compatible with** almost all commercial software. *On the other hand,* Linux gives me more control and better tools for scripting, and usually feels faster on the same hardware.

Step 5. Creative Thinking

Apart from traditional desktop OS, I want to mention real-time operating systems (RTOS) such as FreeRTOS or QNX. *Not only* are they used in cars, medical devices, and industrial robots, *but also* they guarantee that a task will finish within a strict time limit. *Consequently,* I think RTOS are one of the most important — but least visible — types of OS, because they literally keep airplanes flying and cars driving safely.

Step 6. Conclusion

To sum up, an **operating system** connects hardware and applications, and there are many types — GUI, multi-user, multiprocessing, multitasking, and multithreading. *Additionally,* operating systems **control the computer's/system resources**, manage processes, memory, and devices, and provide an interface for programmers and users. The history of OS shows how we moved from mechanical switches and punchcards to modern Windows and Linux. *In conclusion,* as hardware keeps evolving, operating systems will keep changing too.

Blue/Bold = Active Vocabulary, Green/Bold Italic = Linkers

Active Vocabulary: 41, Linkers: 22. Total: 767 words.